AFTER SHRINKING THE OPEN THE SUPPORT OF THE CONSTRUCTED MODULE IS PROPER OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0CTI (sources/stacks-flat/pinned-original-source.tex line 8245).

Allowed standard prerequisites (editorial).

- The pair (A, Ker(A -> A_n)) is henselian. More on Algebra, Lemma labelled more-algebra-lemma-limit-henselian, cited at proof lines 8255-8258. Makes the henselian splitting lemma applicable.
- The finite presentation of F and the identification of the reductions of its support. Same chapter, Lemma labelled lemma-finitely-presented part (3), cited at proof lines 8260-8264. Identifies Z x Spec(A_n) with the support of F_n and shows Z is closed.
- Splitting off the proper part over a henselian pair. More on Morphisms, Lemma labelled more-morphisms-lemma-split-off-proper-part-henselian, cited at proof lines 8266-8270. Produces Z_1 proper over A with the required reductions, leaving Z_2 off the closed fibre.

Source reference. Original reference 0CTC: sources/stacks-flat/pinned-original-source.tex, source lines 8036–8036. Same-chapter pointer into the author environment situation at source line 8036; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CTD: sources/stacks-flat/pinned-original-source.tex, source lines 8062–8062. Same-chapter pointer into the author environment lemma at source line 8062; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CTH: sources/stacks-flat/pinned-original-source.tex, source lines 8181–8181. Same-chapter pointer into the author environment lemma

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at source line 8181; invoked at proof line 8259 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CT7: sources/stacks-chow/tags.txt, source lines 16057–16057. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8255 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CT9: sources/stacks-chow/tags.txt, source lines 16059–16059. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8264 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0CTC let K be as in Lemma 0CTD. Let $W \subset X$ be as in Lemma 0CTH. Set $\mathcal{F} = H^0(K)|_W$. Then, after possibly shrinking the open W, the support of $\mathcal{F}$ is proper over A.*

Proof. Fix $n \geq 1$. Let $I_n = \mathrm{Ker}(A \to A_n)$. By More on Algebra, Lemma 0CT7 the pair (A, I_n) is henselian. Let $Z \subset W$ be the support of $\mathcal{F}$. This is a closed subset as $\mathcal{F}$ is of finite presentation. By part (3) of Lemma 0CTH we see that $Z \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_n)$ is equal to the support of $\mathcal{F}_n$ and hence proper over $\mathrm{Spec}(A/I)$. By More on Morphisms, Lemma 0CT9 we can write $Z = Z_1 \amalg Z_2$ with Z_1, Z_2 open and closed in Z, with Z_1 proper over A, and with $Z_1 \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I_n)$ equal to the support of $\mathcal{F}_n$. In other words, Z_2 does not meet X_0. Hence after replacing W by $W \setminus Z_2$ we obtain the lemma. $\square$